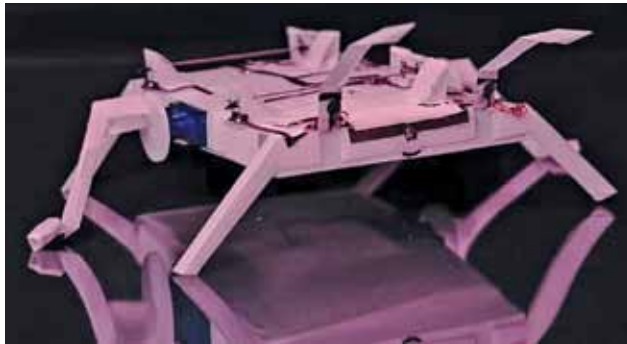




Soft robot jumping without battery shows new robotics approach

Research conducted by a team at the University of California has demonstrated a battery-free soft robot capable of jumping 188 times continuously, highlighting a new approach to energy-efficient robotics. The insect-scale robot



Made from functional sheet materials and threads, the robot detects obstacles and makes decisions to reverse direction (Credit: Wenzhong Yan/UC DAVIS)

operates without motors, electronics, or batteries, using light-responsive liquid crystal elastomers that bend under illumination to store and release elastic energy for repeated jumps. The advancement showed the robot can also carry loads up to 1700 times its own weight. During each leap, its shadow briefly blocks the light source, allowing the material to reset and trigger another jump. The design illustrates 'mechanical intelligence,' where sensing and actuation are embedded directly within the material structure.

Silicon-photonic chip could dramatically reduce energy demands of AI systems

Engineers developed the technology at the University of Florida, where researchers are exploring photonic computing as a scalable path to faster, greener AI hardware for future data centres and edge devices. A silicon-photonic AI chip that performs computations using light rather than electricity could dramatically reduce the energy demands of artificial intelligence systems. The device converts input data into laser light. It processes it through microscopic Fresnel lenses integrated onto silicon, enabling optical convolution, the key pattern-recognition operation in AI, before converting results back into digital signals. Tests showed around 98% accuracy on image-classification tasks while delivering up to 100x higher energy efficiency than conventional electronic chips.

Nano transistor could slash chip power

A team of researchers at Peking University has built a ferroelectric transistor that runs at 0.6 volts with a 1-nanometre gate, potentially cutting power use in chips. Today's logic

chips operate at 0.7V, but most non-volatile memory still needs 5V or more, forcing circuits to step voltage up and down, a process that wastes energy, space, and slows data transfer. In AI chips, moving data eats 60-90% of power. These nano-gate transistors align memory and logic voltages, a step that could reshape how memory and processors work. The principle underlying this technology is universal and can be applied to mainstream ferroelectric materials.

Blockchain 'black box' for drones provides a secure flight activity record

Developed by engineering students at the University of Southampton in the UK, a blockchain-based 'black box' system for drones has been created to provide a secure, tamper-proof record of flight activity, improving transparency and accountability in autonomous machines. The system records opera-



The demonstration represents more than a successful drone test. It signals a future in which intelligent machines do not simply operate but can prove what they have done (Credit: <https://interestingengineering.com>)

tional and sensor data directly onto a decentralised ledger during flight, ensuring the information cannot be altered or deleted. The technology embeds a lightweight blockchain protocol into the drone's microprocessor. Live flight tests showed the system can operate on resource-constrained hardware. Researchers say the approach could help regulators investigate incidents and may extend to autonomous vehicles, industrial robots, and other AI-driven systems.

Solid-state battery with higher discharge capacity even at 100°C

A solid-state battery capable of operating at 100°C while delivering higher discharge capacity has been validated in independent testing, marking progress toward safer high-temperature energy storage. The Doughnut Solid State Battery V1, rated at 26Ah and 94Wh, delivered 27.6Ah at 100°C and 27.5Ah at 80°C, both higher than its room-temperature output. Testing carried out by the VTT Technical Research Centre of Finland confirmed that the battery continued to function normally after the high-temperature trials. Unlike conventional lithium-ion cells, which degrade at extreme temperatures, the solid-state design avoids flammable liquid electrolytes.